COMPOSITION COMPATIBILITY OF THE EXCHANGE MAPS FOR A STACKED PAIR OF PROPER-AND-OPEN SQUARES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Exchange map for proper and separated locally quasi-finite morphisms (Lemma lemma-shriek-proper-and-open, canonical tag 0F7B). cited at statement line 2845 and proof lines 2867-2872, 2884-2888. Supplies the two square exchange maps, and its proof is what shows the representing maps are quasi-isomorphisms.
- K-injectivity of pushforwards along proper morphisms (Cohomology on Sites, Lemma sites-cohomology-lemma-K-injective-flat, canonical tag 093Y). cited at proof lines 2873-2877. Lets Rg_* act as g_* on the K-injective representatives (the author notes this shortcut could be avoided).

Source reference. Original lemma 0F7B: sources/stacks-chow/tags.txt, source lines 19001–19001. Cited at statement line 2845 and proof lines 2867-2872, 2884-2888: supplies the two square exchange maps and the proof that their representing maps are quasi-isomorphisms.

Source reference. Original lemma 093Y: sources/stacks-chow/tags.txt, source lines 11531–11531. Cited at proof lines 2873-2877: the pushed-forward complexes are K-injective, so the derived pushforward acts as the plain pushforward on them.

Lemma 0.2. *Consider a commutative diagram of schemes*

$$\begin{array}{ccc} X' & \xrightarrow{k} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{l} & Y \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ Z' & \xrightarrow{m} & Z \end{array}$$

with f, f', g and g' proper and k, l, and m separated and locally quasi-finite. Then the isomorphisms of Lemma 0F7B for the two squares compose to give the isomorphism for the outer rectangle (see proof for a precise statement).

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Proof. The statement means that if we write $R(g \circ f)_* = Rg_* \circ Rf_*$ and $R(g' \circ f')_* = Rg'_* \circ Rf'_*$, then the isomorphism $m_! \circ Rg'_* \circ Rf'_* \to Rg_* \circ Rf_* \circ k_!$ of the outer rectangle is equal to the composition

$$m_! \circ Rg'_* \circ Rf'_* \to Rg_* \circ l_! \circ Rf'_* \to Rg_* \circ Rf_* \circ k_!$$

of the two maps of the squares in the diagram. To prove this choose a K-injective complex $\mathcal{J}^\bullet$ of Λ-modules on $X'_{étale}$ and a quasi-isomorphism $k_!\mathcal{J}^\bullet \to \mathcal{I}^\bullet$ to a K-injective complex $\mathcal{I}^\bullet$ of Λ-modules on $X_{étale}$. The proof of Lemma 0F7B shows that the canonical map

$$a : l_! f'_* \mathcal{J}^\bullet \to f_* \mathcal{I}^\bullet$$

is a quasi-isomorphism and this quasi-isomorphism produces the second arrow on applying Rg_*. By Cohomology on Sites, Lemma 093Y the complex $f_*\mathcal{I}^\bullet$, resp. $f'_*\mathcal{J}^\bullet$ is a K-injective complex of Λ-modules on $Y_{étale}$, resp. $Y'_{étale}$. (Using this is cheating and could be avoided.) In particular, the same reasoning gives that the canonical map

$$b : m_! g'_* f'_* \mathcal{J}^\bullet \to g_* f_* \mathcal{I}^\bullet$$

is a quasi-isomorphism and this quasi-isomorphism represents the first arrow. Finally, the proof of Lemma 0F7B show that $g_* l_! f'_! \mathcal{J}^\bullet$ represents $Rg_*(l_! f'_* \mathcal{J}^\bullet)$ because $f'_* \mathcal{J}^\bullet$ is K-injective. Hence $Rg_*(a) = g_*(a)$ and the composition $g_*(a) \circ b$ is the arrow of Lemma 0F7B for the rectangle. $\square$